Recursivity and the Estimation of Dynamic Games with Continuous Controls

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Introduction

Many questions in IO involve firm choices with **persistent effects on market structure**.

- Examples: R&D, productive capacity, product characteristics, advertising ...
- Dynamic; often in concentrated markets; naturally continuous.

$\Rightarrow$ **Dynamic Games with Continuous Controls** (DGCCs).

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$\Rightarrow$ **Dynamic Games with Continuous Controls** (DGCCs).

We study their estimation. Two observations:

- If observed policies are a **MPE**, they must satisfy firms' Bellman equations.
- $\widehat{\text{Transitions}} + \widehat{\text{Continuation}}(\boldsymbol{\theta}) \Rightarrow$ can form "empirical" Bellman equations.

This Paper

We **propose** estimators that exploit the empirical Bellman (*Recursive Estimators* (REs)).

- Models with shocks to firms' investment cost $\Rightarrow$ Pseudo-MLE.
- Models without such shocks $\Rightarrow$ NLLS.

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We **evaluate** REs relative to Bajari, Benkard, and Levin (2007) (BBL).

- $V(\text{State} \mid \textbf{Observed Policy}) \geq V(\text{State} \mid \textbf{Deviation}) \ \forall$ Deviation.
- **Q**: Which deviations? We deploy multiple implementations.
- Monte Carlos based on Bajari et al. (2007) and Hashmi and van Biesebroeck (2016).

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Findings

1. REs substantially outperform BBL.
   - Pseudo-MLE recovers investment costs with small bias.
   - All three BBL variants fail: wrong sign, at the boundary, or orders of magnitude off.
   - Entry cost bias $\sim 5\%$ (BBL: 30–70%) and $\sim 2\times$ more precise; scrap values comparable.
2. BBL performance depends markedly on the deviation choice.
   - The ranking of deviations flips across environments.

Related Literature and Contributions

Estimation of DGCCs

- Bajari et al. (2007), Jofre-Bonet and Pesendorfer (2003), Schrimpf (2011), **Srisuma (2013)**.
- Applications: Ryan (2012), Hashmi and van Biesebroeck (2016), Fowlie, Reguant, and Ryan (2016), Liu and Siebert (2022), Egan, Hortaçsu, Kaplan, Sunderam, and Yao (2025), **Wang and Zhai (2025)**.

Estimation of Dynamic Discrete Games

- Aguirregabiria and Mira (2007), Pakes, Ostrovsky, and Berry (2007), Pesendorfer and Schmidt-Dengler (2008).

Contributions.

- Derivation of estimators for DGCCs based on recursive equilibrium conditions.
  - Sufficient conditions for policy monotonicity (pseudo-MLE).
  - Computationally tractable in empirically realistic settings.
- REs substantially outperform BBL.
  - BBL performance sensitive to the deviation choice.
- Existence of MPEs in DGCCs with shocks to firms' investment cost.

Roadmap

- A **model** of entry, exit, and quality upgrading.
- Introduce estimators based on recursive equilibrium conditions.
- A review of the Bajari et al. (2007) inequality estimator.
- Monte Carlo simulations.

Market Structure and Profits

A variant of Pakes and McGuire (1994)

$\bar{N}$ **single-product firms** in the market

- N active firms, $N \leq \bar{N}$; N endogenous.
- $\bar{N} - N$ potential entrants.

Firms are described by a **characteristic** $\xi_j \in \Xi \subset \mathbb{R} \cup \{-\infty\}$

- Quality, capacity, technology, consumer "goodwill", ...
- $\xi_j = -\infty$ indicates being inactive.

The **industry state** is $\boldsymbol{\xi} = (\xi_1, \ldots, \xi_{\bar{N}})$.

Firms earn **flow profits** that depend on their qualities: $\pi_j(\boldsymbol{\xi})$.

- $\pi_j(\boldsymbol{\xi})$ are *symmetric* – Doraszelski and Satterthwaite (2010):

Quality Transitions

Firms can invest to affect the evolution of their qualities.

Assumption (Conditional Independence and No Spillovers (CINS))

$$F(\boldsymbol{\xi}' \mid \boldsymbol{\xi}, \boldsymbol{x}) = \prod_{j=1}^{\bar{N}} F(\xi_j' \mid \xi_j, x_j) \ .$$

- Straightforward to accommodate an aggregate shock.
- **Satisfied by**: capacity accumulation (Besanko and Doraszelski (2004), Ryan (2012)); quality ladders (Doraszelski and Pakes (2007), Hashmi and van Biesebroeck (2016)).
- **Ruled out**: demand with switching costs (Egan et al. (2025)) – state is lagged market shares, which depend on all firms' prices.

Investment Costs

Investment is costly: $c(x, \nu)$

- $\nu \sim F_\nu$ is a private **investment cost shock**.
  - F_ν is *known* to the econometrician.

Assumption (Investment Cost (IC))

$c(x, \nu)$ is twice continuously differentiable, with

$$\frac{\partial c}{\partial x} > 0, \quad \frac{\partial^2 c}{\partial x^2} \geq 0, \quad \frac{\partial^2 c}{\partial \nu \partial x} > 0 \, .$$

Timing and Solution Concept

1	2	3	4	5	6
Incumbents earn $\pi_j(\boldsymbol{\xi})$	$\rho \sim F_\rho$ $\phi \sim F_\phi$	Entry/exit decisions	$\nu \sim F_\nu$ (active firms)	Investment decisions	Transitions $\xi' \sim F(\xi' \mid \xi, x)$

Definition (Symmetric MPE)

A **Symmetric Markov Perfect Equilibrium** (SMPE) is a tuple

$$\sigma(\boldsymbol{\xi}, \varepsilon) = (\alpha^E(\boldsymbol{\xi}, \phi), \alpha^I(\boldsymbol{\xi}, \rho), \sigma^x(\boldsymbol{\xi}, \nu))$$

such that, given that all firms behave according to this tuple,

1. $\alpha^E(\boldsymbol{\xi}, \phi) \in \{0, 1\}$ maximizes the potential entrant's ENPV of profits;
2. $\alpha^I(\boldsymbol{\xi}, \rho) \in \{0, 1\}$ maximize the incumbent's ENPV of profits.
3. $\sigma^x(\boldsymbol{\xi}, \nu)$ maximizes the ENPV of profits of a firm that chooses to be active in $t + 1$.

The Investment Problem

The firm's investment problem can be written as

$$\max_{x \in \mathbb{R}_+} \left\{ -c(x,\nu) + \beta \int_{\xi_1'} W(\xi_1' \mid \boldsymbol{\xi}, F^\sigma) \, \mathrm{d}F(\xi_1' \mid \xi_1, x) \right\},$$

where

- $W(\xi_1' \mid \boldsymbol{\xi}, F^\sigma) := \int_{\xi_2'} \cdots \int_{\xi_{\bar{N}}'} \bar{V}_I(\xi_1', \boldsymbol{\xi}_{-1}') \, \mathrm{d}F^\sigma(\xi_{\bar{N}}' \mid \boldsymbol{\xi}_{\bar{N}}) \ldots \mathrm{d}F^\sigma(\xi_2' \mid \boldsymbol{\xi}_2)$;
- $\bar{V}_I(\boldsymbol{\xi}) := \int_\rho V_I(\boldsymbol{\xi}, \rho) \, \mathrm{d}F_\rho$.
- $F^\sigma(\xi_j' \mid \boldsymbol{\xi}_j) := \int_{\varepsilon_j} F(\xi_j' \mid \xi_j, \sigma_j(\boldsymbol{\xi}_j, \varepsilon_j)) \, \mathrm{d}G_{\varepsilon_j}$;

We would like the problem above to have a **unique solution, monotonic in** ν.

- Computing equilibria.
- Estimation.

Unique Optimal Investments

Definition (Doraszelski and Satterthwaite (2010))

The distribution $F(\xi' \mid \xi, x)$ is UIC-admissible if, for all $\xi', \xi \in \Xi$, and $x \in \mathbb{R}_+$

$$F(\xi' \mid \xi, x) = L(\xi', \xi) + K(\xi', \xi) Q(\xi, x) \ ,$$

where $Q(\xi, x)$ is $\mathcal{C}^2$ in x for all $\xi \in \Xi$ and satisfies

$$\frac{\partial Q(\xi, x)}{\partial x} > 0 \quad \text{and} \quad \frac{\partial^2 Q(\xi, x)}{\partial x^2} < 0 \ .$$

Proposition

If Assumption IC holds and $F(\xi' \mid \xi, x)$ is **UIC-admissible**, *then for all $(\boldsymbol{\xi}, \nu) \in \Xi \times \mathrm{supp}(F_\nu)$ the investment problem has a* **unique solution** *$\sigma^x(\boldsymbol{\xi}, \nu)$. Moreover, $\sigma^x(\boldsymbol{\xi}, \nu)$ is* **increasing in** *ν, strictly so over the range where it is interior.*

▸ Alternative conditions for uniqueness.

Existence and Equilibrium Computation

Proposition

Suppose $|\Xi| < \infty$. Then there exists a Symmetric MPE.

- Not previously established for this class of models.
- Despite the continuous controls, good old Brouwer suffices.
  - Only endogenous parameter in the investment problem is F^σ, **finite-dimensional**.
  - We prove the existence of a fixed point for F^σ and from it construct an equilibrium.
  - Akin to the *Representation Lemma* in Aguirregabiria and Mira (2007).

Computation

- Looks for a fixed point of $(F^\sigma, \bar{V}_I)$.
- Not dissimilar to Pakes and McGuire (1994).

Roadmap

- A model of entry, exit, and quality upgrading.
- Introduce estimators based on **recursive equilibrium conditions**.
- A review of the Bajari et al. (2007) inequality estimator.
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The Estimation Problem

The econometrician observes a panel of M markets over T periods

- For each of the N_{mt} active firms, the data include
  - the firm's characteristic ξ_{jmt};
  - the firm's investment x_{jmt};

Flow profits and transitions

- Can be estimated "offline".
  - I.e., without imposing dynamic equilibrium conditions.

What remains to be estimated

- Investment cost parameters $\boldsymbol{\theta}_x$ in $c(x, \nu; \boldsymbol{\theta}_x)$.
- Scrap value parameters $\boldsymbol{\theta}_\rho$ in $F_\rho(\rho; \boldsymbol{\theta}_\rho)$.
- Entry cost parameters $\boldsymbol{\theta}_\phi$ in $F_\phi(\phi; \boldsymbol{\theta}_\phi)$.

Recursive Estimators

- Assume (as in BBL) we have estimates of
  - $\hat{F}(\xi' \mid \xi, x)$
  - $\hat{F}^{\sigma}(\xi' \mid \boldsymbol{\xi})$
  - $\widehat{\bar{V}_I}(\boldsymbol{\xi}; \boldsymbol{\theta}_{-\phi})$ of $\bar{V}_I(\boldsymbol{\xi}; \sigma, \boldsymbol{\theta}_{-\phi})$, with $\boldsymbol{\theta}_{-\phi} := (\boldsymbol{\theta}_x, \boldsymbol{\theta}_\rho)$.
    - $\rightarrow$ Relies on an estimate $\hat{\sigma}^x(\boldsymbol{\xi}, \nu)$.
    - $\rightarrow$ We discuss below how to estimate $\bar{V}_I$ and σ^x.

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Then we can set up firms' **empirical Bellman equations**:

$$\max_{x \in \mathbb{R}_+} \left\{ -c(x, \nu; \boldsymbol{\theta}_x) + \beta \int_{\xi_1'} \widehat{W}(\xi_1' \mid \boldsymbol{\xi}; \hat{F}^{\sigma}, \boldsymbol{\theta}_{-\phi}) \, \mathrm{d}\hat{F}(\xi_1' \mid \xi_1, x) \right\}$$

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We exploit the empirical Bellman equations and their FOCs to construct estimators of $\boldsymbol{\theta}$.

Recursive Estimators: Pseudo-MLE

<u>Investment contribution</u>: conditional CDF

$$F_X(x \mid \boldsymbol{\xi}; \boldsymbol{\theta}_{-\phi}, \hat{\Phi}) = 1 - F_\nu\Big((\sigma^x)^{-1}(x; \boldsymbol{\xi}; \boldsymbol{\theta}_{-\phi}, \hat{\Phi})\Big)$$

- $x = 0$: contributes mass $F_X(0 \mid \boldsymbol{\xi})$.
  - **Empirical Bellman** FOC contributes $(\sigma^x)^{-1}$.
- $x > 0$: compute density $f_X(x \mid \boldsymbol{\xi}; \boldsymbol{\theta}_{-\phi}, \hat{\Phi})$.
  - **Empirical Bellman** FOC contributes $\frac{d}{dx}(\sigma^x)^{-1}$.

Recursive Estimators: Pseudo-MLE

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<u>Active-status contribution</u> (entry analogous):

$$\mathbb{P}(\text{Active} \mid \boldsymbol{\xi}; \boldsymbol{\theta}) = F_\rho\left(\widehat{\bar{V}}_I^A(\boldsymbol{\xi}; \boldsymbol{\theta}_{-\phi}) - \pi(\boldsymbol{\xi}); \boldsymbol{\theta}_\rho\right)$$

- **Empirical Bellman** used to obtain the estimate $\widehat{\bar{V}}_I^A(\boldsymbol{\xi}; \boldsymbol{\theta}_{-\phi})$.

Recursive Estimators: Pseudo-MLE

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- **Empirical Bellman** used to obtain the estimate $\widehat{\bar{V}}_I^A(\boldsymbol{\xi}; \boldsymbol{\theta}_{-\phi})$.

$\Rightarrow$ Likelihood $L(\boldsymbol{\theta}) = \prod_{f,m} L_X(\boldsymbol{\theta}; x_{fm}, \boldsymbol{\xi}_{fm}) \cdot L_A(\boldsymbol{\theta}; a_{fm}, \boldsymbol{\xi}_{fm})$.

How to Estimate Continuation Values?

We show that

$$[I - \beta \boldsymbol{M}(\boldsymbol{P})]\bar{\boldsymbol{V}}_I = \boldsymbol{\pi} - \boldsymbol{K}(\boldsymbol{\theta}_x) + \boldsymbol{\Sigma}(\boldsymbol{\theta}_\rho)\ . \tag{1}$$

Details

- $\bar{V}_I$ estimable up to $\boldsymbol{\theta}$; no simulation error.
- Similar calculations in Jofre-Bonet and Pesendorfer (2003) and Pakes et al. (2007).

Is it feasible to repeatedly solve (1)**?**

- Some models have special structure that makes it very fast (more below).
- O.w., use e.g. Chen (2025).
- Alternatively, simulate $W(\xi' \mid \boldsymbol{\xi}; \boldsymbol{\theta}_x, \boldsymbol{\theta}_\rho)$

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Bajari et al. (2007)

If σ is a SMPE then $\forall\, \boldsymbol{\xi}, \sigma'$

$$\bar{V}_I(\boldsymbol{\xi}; \sigma, \sigma, \boldsymbol{\theta}_x, \boldsymbol{\theta}_\rho) \geq \bar{V}_I(\boldsymbol{\xi}; \sigma', \sigma, \boldsymbol{\theta}_x, \boldsymbol{\theta}_\rho);$$

- Analogous condition holds for $\bar{V}_E$.

Define

$$g_s(\boldsymbol{\xi}, \sigma'; \sigma, \boldsymbol{\theta}) := \bar{V}_s(\boldsymbol{\xi}; \sigma, \sigma, \boldsymbol{\theta}) - \bar{V}_s(\boldsymbol{\xi}; \sigma', \sigma, \boldsymbol{\theta}) \,, \quad s \in \{I, E\} \,.$$

The **BBL estimator** is

$$\hat{\boldsymbol{\theta}}_{BBL} = \arg\min_{\boldsymbol{\theta}} \hat{Q}(\boldsymbol{\theta}, \hat{\sigma}) = \frac{1}{n_I} \sum_{i=1}^{n_I} \left(\min \left\{ \hat{g}_{s_i}(\boldsymbol{\xi}_i, \sigma_i'; \hat{\sigma}, \boldsymbol{\theta}), 0 \right\} \right)^2$$

- Estimates are a function of the chosen **deviations** σ'.
  - How to choose deviations σ'?
  - Choice does matter: Srisuma (2013) and simulations below.

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Monte Carlo Setup: BBL Design

We specialize the framework above.

- A **Bajari et al. (2007)**-inspired design with entry and exit.
- $\Xi = \{-\infty, -\ln 20, -\ln 19, \ldots, \ln 19, \ln 20\}$.
- $\pi(\boldsymbol{\xi})$ is derived from
  - Single-product firms facing simple logit demand.
  - Constant marginal cost μ.
  - Nash-Bertrand competition.
  - ▸ Flow Profit Derivation

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  - ▸ Flow Profit Derivation
- Transitions
  - $\xi' \in \{\xi^-, \xi, \xi^+\}$; independent positive and negative shocks.
  - Negative shock: $\mathbb{P}(\xi' = \xi^-) = \delta$.
  - Positive shock: $\mathbb{P}(\xi' = \xi^+) = u(x) = \frac{\psi x}{1+\psi x}$.
  - ▸ Computational payoff of local transitions + absorbing exit

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  - ▸ Computational payoff of local transitions + absorbing exit
- $c(x) = \kappa x$; no investment cost shock.

Monte Carlo Setup: BBL Design — Parameters

Parameter	Value
Data Structure	
Number of Quality Levels	39
Minimum Quality Level	$-\ln 20$
Maximum Quality Level	$\ln 20$
Maximum Number of Players	3
Number of Markets	100
Number of Periods	40
Model Parameters	
Discount Factor (β)	0.925
Demand System (γ_0, α)	[0.1, -0.25]
Marginal Cost (θ_c)	[1.09861, 0.0]
Quality Transition (δ, ψ)	[0.7, 7.0]
Investment Cost (κ)	1.0
Investment Cost Shock Distribution (F_ν)	Dirac(0.0)
Scrap Value Distribution (F_ρ)	Uniform(22.0, 23.0)
Entry Costs Distribution (F_ϕ)	Uniform(22.0, 30.0)
Forward Simulation Settings	
Number of Simulated Paths	250
Simulation Horizon	150

First Stage: BBL Design

- $\pi(\boldsymbol{\xi})$
  - Treated as known.
- $P(\xi' \mid \xi, x)$
  - MLE, parametric structure imposed above.
- $\sigma^x(\boldsymbol{\xi})$
  - Degenerate $F_\nu \Rightarrow$ investment is deterministic in $\boldsymbol{\xi}$.
  - OLS regression of x on functions of $\boldsymbol{\xi}$.
- $P^\sigma(\xi' \mid \boldsymbol{\xi}) = P(\xi' \mid \xi, \sigma^x(\boldsymbol{\xi}))$
  - Combine previous two steps.
- Probabilities of entry and exit.
  - Logit on functions of $\boldsymbol{\xi}$.

Monte Carlo Results: BBL Design

Parameter	Value	Recursive	Additive	Multiplicative	Asymptotic
κ	1.000	1.001	0.633	0.542	-4.214
		(0.009)	(0.026)	(0.058)	(2.369)
F_ρ Lower Bound	22.000	21.756	20.372	18.848	22.397
		(0.439)	(1.157)	(1.036)	(3.999)
F_ρ Upper Bound	23.000	22.967	23.614	22.319	27.185
		(0.048)	(0.126)	(0.110)	(3.287)
F_ϕ Lower Bound	22.000	21.487	22.373	22.090	779.514
		(0.746)	(1.639)	(1.195)	(3504.104)
F_ϕ Upper Bound	30.000	32.314	33.742	27.087	1196.404
		(3.788)	(136.493)	(13.419)	(4953.821)

- "Recursive" = NLLS on recursive equilibrium conditions.
  - Degenerate $F_\nu \Rightarrow$ pseudo-MLE inapplicable.

Policy and Deviation Details

Monte Carlo Setup: HvB Design

- Adding entry and exit to the **Hashmi and van Biesebroeck (2016)** model.

- $\Xi = \{-\infty, \xi_{\min}, \xi_{\min} + \delta, \ldots, \xi_{\max} - \delta, \xi_{\max}\}$.
- $\pi(\boldsymbol{\xi})$ is derived from
  - Single-product firms facing nested logit demand.
  - Constant marginal cost $mc(\xi)$.
  - Nash-Bertrand competition.
  - Flow Profit Derivation
- Transitions
  - $\xi' \in \{\xi^-, \xi, \xi^+\}$; independent positive and negative shocks.
  - Negative shock: $\mathbb{P}(\xi' = \xi^-) = \theta_{t1}$.
  - Positive shock: $\mathbb{P}(\xi' = \xi^+ \mid \xi, x) = 1 - (1 + x)^{-\lambda(\xi)}$.
  - $\lambda(\xi) = \exp(\theta_{t2} + \theta_{t3}\xi + \theta_{t4}\xi^2)$: decreasing in ξ.
- $c(x, \nu) = \theta_{x1}x + \theta_{x2}x^2 + \theta_{x3}\nu x$.
  - $\nu \sim N(0, 1)$, known to the econometrician.

Monte Carlo Setup: Parameters

Parameter	Value
Data Structure	
Number of Quality Levels	15
Minimum Quality Level	-1.4
Maximum Quality Level	1.4
Distance Across Quality Levels (δ)	0.2
Maximum Number of Players	5
Number of Markets	100
Number of Periods	40
Model Parameters	
Discount Factor (β)	0.95
Demand System (γ_0, α)	[1.0, -0.222]
Marginal Cost (θ_c)	[2.47, 0.0]
Quality Transition (θ_t)	[0.347, -0.75, -0.3, -0.1]
Investment Cost (θ_x)	[2.625, 1.624, 0.5096]
Investment Cost Shock Distribution (F_ν)	Normal(0.0, 1.0)
Scrap Value Distribution (F_ρ)	Exponential(0.8)
Entry Costs Distribution (F_ϕ)	Exponential(11.0)
Forward Simulation Settings	
Number of Simulated Paths	250
Simulation Horizon	150

First Stage: HvB Design

- $\boldsymbol{\pi}(\boldsymbol{\xi})$, $P(\xi' \mid \xi, x)$, and entry/exit probabilities: as in the BBL design. ▸ BBL First Stage
- $\sigma^x(\boldsymbol{\xi}, \nu)$
  - $\sigma^x(\boldsymbol{\xi}, \nu) = F_X^{-1}(1 - F_\nu(\nu) \mid \boldsymbol{\xi})$.
  - We estimate evenly spaced **quantile regressions** of investment on functions of $\boldsymbol{\xi}$.
  - Commonly used $\mathbb{E}[x \mid \boldsymbol{\xi}]$ ignores dependence on ν.
- $P^\sigma(\xi' \mid \boldsymbol{\xi}) \approx \sum_k \omega_k P(\xi' \mid \xi, \sigma(\boldsymbol{\xi}, \nu_k))$
  - Combine $\hat{\sigma}^x$ with the estimated quality transitions.

Monte Carlo Results

Parameter	Value	Recursive	Additive	Multiplicative	Asymptotic
θ_{x1}	2.625	2.920	-4.972	-5.000	-4.829
		(0.261)	(0.360)	(0.000)	(1.069)
θ_{x2}	1.624	1.603	-5.000	-4.763	2.745
		(0.068)	(0.000)	(0.983)	(2.533)
θ_{x3}	0.510	0.548	9.324	8.733	0.232
		(0.023)	(2.424)	(3.259)	(1.420)
F_ρ Scale Parameter	0.800	0.754	0.901	0.702	0.721
		(0.061)	(0.031)	(0.079)	(0.035)
F_ϕ Scale Parameter	11.000	10.427	18.976	17.514	14.311
		(0.788)	(1.631)	(2.224)	(1.641)

- "Recursive" = pseudo-MLE on recursive equilibrium conditions.

Policy and Deviation Details

Conclusion

Motivation

Many IO decisions have **persistent effects** on market structure and are **naturally continuous**.

Yet DGCC applications are **scarce**.

Why? Estimator performance, we hypothesize.

Propose

Estimators from **recursive equilibrium conditions** (REs):

- **Pseudo-MLE** (cost shocks)
- **NLLS** (no shocks)

En route: policy monotonicity + SMPE existence.

Find

In our MC:

- REs substantially **outperform BBL**.
- All three BBL variants **fail**: wrong sign, at the boundary, or orders off.
- Deviation ranking **changes across designs**.

Future → multiple continuous controls; more general state transitions.

Thank you!

Value Functions: The Incumbent's Investment Problem

The expected net present value (ENPV) to the incumbent entering the period is

$$\bar{V}_I(\xi) = \int_\rho V_I(\xi, \rho) dF_\rho = \int_\rho \max\left\{\pi(\xi) + \rho, \int V_I^A(\xi, \nu) dF_\nu\right\} dF_\rho$$

where $V_I^A(\xi, \nu)$ is the ENPV of having chosen to be active next period, i.e.

$$V_I^A(\xi, \nu) = \max_{x \in \mathbb{R}_+} \left\{\pi(\xi) - c(x, \nu) + \beta \mathbb{E}[V_I(\xi', \rho) \mid \xi, x, \sigma]\right\} \qquad \text{(Investment)}$$

and the continuation value is

$$\mathbb{E}[V_I(\xi', \rho) \mid \xi, x, \sigma] = \int_{\varepsilon_{-1}} \int_{\xi'} \bar{V}_I(\xi') \mathrm{d}F(\xi' \mid \xi, x, \sigma_{-1}(\xi, \varepsilon_{-1})) \mathrm{d}G_{\varepsilon_{-1}}$$

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and the continuation value is , by indep. of ε_j

$$\mathbb{E}[V_I(\boldsymbol{\xi}', \rho) \mid \boldsymbol{\xi}, x, \sigma] = \int_{\varepsilon_2} \cdots \int_{\varepsilon_{\bar{N}}} \int_{\boldsymbol{\xi}'} \bar{V}_I(\boldsymbol{\xi}') \mathrm{d}F(\boldsymbol{\xi}' \mid \boldsymbol{\xi}, x, \sigma_{-1}(\boldsymbol{\xi}, \boldsymbol{\varepsilon}_{-1})) \mathrm{d}G_{\varepsilon_{\bar{N}}} \ldots \mathrm{d}G_{\varepsilon_2}$$

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and the continuation value is , by indep. of ε_j and by conditional indep. of $\boldsymbol{\xi}' \mid \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{x}$

$$\mathbb{E}[V_I(\boldsymbol{\xi}', \rho) \mid \boldsymbol{\xi}, x, \sigma] = \int_{\xi_1'} \int_{\xi_{\bar{N}}'} \cdots \int_{\xi_2'} \bar{V}_I(\boldsymbol{\xi}') \prod_{j=2}^{\bar{N}} \int_{\varepsilon_j} \mathrm{d}F(\xi_j' \mid \xi_j, \sigma_j(\boldsymbol{\xi}_j, \varepsilon_j)) \mathrm{d}G_{\varepsilon_j} \mathrm{d}F(\xi_1' \mid \xi_1, x)$$

Value Functions: The Incumbent's Investment Problem

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Value Functions: The Incumbent's Investment Problem

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and the continuation value is , by indep. of ε_j and by conditional indep. of $\boldsymbol{\xi}' \mid \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{x}$

$$\mathbb{E}[V_I(\boldsymbol{\xi}', \rho) \mid \boldsymbol{\xi}, x, \sigma] = \int_{\xi_1'} W(\xi_1' \mid \boldsymbol{\xi}; F^\sigma) \mathrm{d}F(\xi_1' \mid \xi_1, x) \qquad \text{(Continuation)}$$

$$W(\xi_1' \mid \boldsymbol{\xi}; F^\sigma) = \int_{\xi_{\bar{N}}'} \cdots \int_{\xi_2'} \bar{V}_I(\xi_1', \boldsymbol{\xi}_{-1}') \mathrm{d}F^\sigma(\xi_2' \mid \boldsymbol{\xi}_2) \ldots \mathrm{d}F^\sigma(\xi_{\bar{N}}' \mid \boldsymbol{\xi}_{\bar{N}})$$

Value Functions: The Incumbent's Investment Problem

Value Functions: Exit

Incumbents must commit to exiting before knowing ν. Their ENPV given ρ is

$$V_I(\xi, \rho) = \max\left\{\pi(\xi) + \rho, \int V_I^A(\xi, \nu) dF_\nu\right\} \tag{Exit}$$

Hence the conditional probability of exiting is

$$\mathbb{P}(\alpha^I(\xi, \rho) = 1 | \xi) = 1 - F_\rho\left(\int V_I^A(\xi, \nu) dF_\nu - \pi(\xi)\right)$$

Value Functions: Entry

Entrants are short-lived: they either enter or perish. Their ENPV given ϕ is

$$V_E(\boldsymbol{\xi}_{-1}, \phi) = \max\left\{0, \bar{V}_E^A(\boldsymbol{\xi}_{-1}) - \phi\right\} \tag{Entry}$$

where

$$\bar{V}_E^A(\boldsymbol{\xi}_{-1}) = \int_\nu \max_{x \in \mathbb{R}_+} \left\{ -c(x,\nu) + \beta \int_{\xi_1'} W(\xi_1'|(-\infty, \boldsymbol{\xi}_{-1}), F^\sigma) \mathrm{d}F(\xi_1'|\xi_E, x) \right\} \mathrm{d}F_\nu$$

where $\xi_E \in \Xi$ is an exogenously specified quality for the entrant.
Hence the conditional probability of entering is

$$\mathbb{P}\big(\alpha^E(\boldsymbol{\xi}_{-1}, \phi) = 1 | \boldsymbol{\xi}_{-1}\big) = F_\phi\big(\bar{V}_E^A(\boldsymbol{\xi}_{-1})\big)$$

Back to investment problem.

Investment: First Order Condition

Consider firm 1's problem (wlog given symmetry and anonymity). Then

$$-\partial_x c(x,\nu) + \beta\partial_x\left(\int_{\xi_1'} W(\xi_1' \mid \boldsymbol{\xi}, F^\sigma) dF(\xi_1' \mid \xi_1, x)\right) \leq 0 \tag{2}$$

where recall

$$W(\xi_1' \mid \boldsymbol{\xi}; F^\sigma) := \int_{\xi_2'} \cdots \int_{\xi_{\bar{N}}'} \bar{V}_I(\xi_1', \boldsymbol{\xi}_{-1}') \mathrm{d}F^\sigma(\xi_{\bar{N}}' \mid \boldsymbol{\xi}_{\bar{N}}) \ldots \mathrm{d}F^\sigma(\xi_2' \mid \boldsymbol{\xi}_2)$$

and

$$\bar{V}_I(\boldsymbol{\xi}) = \int_\rho V_I(\boldsymbol{\xi}, \rho) dF_\rho = \int_\rho \max\left\{\pi(\boldsymbol{\xi}) + \rho, \int V_I^A(\boldsymbol{\xi}, \nu) dF_\nu\right\} dF_\rho$$

and

$$V_I^A(\boldsymbol{\xi}, \nu) = \max_{x \in \mathbb{R}_+}\left\{\pi(\boldsymbol{\xi}) - c(x,\nu) + \beta\mathbb{E}[V_I(\boldsymbol{\xi}', \rho) \mid \boldsymbol{\xi}, x, \sigma]\right\}$$

Back to investment problem.

Assumption (A1)

Let Ξ be a compact subset of the real line, and denote its minimum and maximum by ξ_m and ξ_M. Let Ξ° be the interior of Ξ. The family of distributions $F(\cdot \mid \xi, x)$ is such that

(a) *$F(\xi' \mid \xi, x)$ is strictly decreasing and strictly convex in x, for all $\xi \in \Xi$ and $\xi' \in \Xi^\circ$;*

(b) *$F(\xi_m \mid \xi, x)$ is decreasing and convex in x, , for all $\xi \in \Xi$.*

Proposition

Assume

(a) *$c(x, \nu)$ is convex in x;*

(b) *$F(\xi' \mid \xi, x)$ is twice continuously differentiable in x for all ξ', ξ;*

(c) *$W(\xi' \mid \boldsymbol{\xi}, F^\sigma)$ is increasing in ξ' for all $\boldsymbol{\xi}$.*

(d) *A1 holds.*

Then for all $(\boldsymbol{\xi}, \nu)$ the investment problem has a **unique solution** *$\sigma^x(\boldsymbol{\xi}, \nu)$. Moreover, $\sigma^x(\boldsymbol{\xi}, \nu)$ is* **increasing in** *ν, strictly so over the range where it is interior.*

Computing Markov Perfect Equilibria

1. Guess $\bar{V}_I(\boldsymbol{\xi})$ and $F^\sigma(\xi' \mid \boldsymbol{\xi})$.
2. Solve incumbents' investment problem at all $\boldsymbol{\xi}$ and judiciously chosen ν.
   - Gives $\bar{V}_I^A := \int V_I^A(\boldsymbol{\xi}, \nu)\, \mathrm{d}F_\nu$.
3. Compute incumbents' optimal exit decisions.
4. Use incumbents' optimal choices to update $\bar{V}_I(\boldsymbol{\xi})$.
5. Solve potential entrants' investment problems at all $\boldsymbol{\xi}_{-1}$ and judiciously chosen ν to obtain $\bar{V}_E^A := \int V_E^A(\boldsymbol{\xi}_{-1}, \nu)\, \mathrm{d}F_\nu$.
6. Compute potential entrants' optimal entry decisions.
7. Iterate to convergence in $\bar{V}_I$ and F^σ.

N.B.:

- We need only compute $\bar{V}_I$.
  - Potential entrants' decisions depend on $\bar{V}_I$.
  - There is no continuation value conditional on not entering: short-lived PEs assumption.
  - Symmetry reduces the computational burden significantly: Pakes and McGuire (1994).

Back to investment problem.

Integrated Value Function

$$[I - \beta \boldsymbol{M}(\boldsymbol{P})]\bar{\boldsymbol{V}}_I = \boldsymbol{\pi} - \boldsymbol{K}(\boldsymbol{\theta}_x) + \boldsymbol{\Sigma}(\boldsymbol{\theta}_\rho)\ .$$

where

$$\boldsymbol{K}(\boldsymbol{\theta}_x) = \left[\mathbb{P}\big(\alpha^I(\boldsymbol{\xi},\rho) = 1 \mid \boldsymbol{\xi}\big) \int c(\sigma^x(\boldsymbol{\xi},\nu),\nu;\boldsymbol{\theta}_x)\, \mathrm{d}F_\nu\right]_{\{\boldsymbol{\xi}\in\boldsymbol{\Xi}_I\}}$$

$$\boldsymbol{\Sigma}(\boldsymbol{\theta}_\rho) = \left[\big[1 - \mathbb{P}\big(\alpha^I(\boldsymbol{\xi},\rho) = 1 \mid \boldsymbol{\xi}\big)\big]\mathbb{E}\big[\rho \mid F_\rho(\rho) > \mathbb{P}\big(\alpha^I(\boldsymbol{\xi},\rho) = 1 \mid \boldsymbol{\xi}\big)\big]\right]_{\{\boldsymbol{\xi}\in\boldsymbol{\Xi}_I\}}$$

$$\boldsymbol{M}(\boldsymbol{P}) = \left[\prod_{j=1}^{\bar{N}} F^\sigma(\xi'_j \mid \boldsymbol{\xi}) : \boldsymbol{\xi}', \boldsymbol{\xi} \in \boldsymbol{\Xi}_I\right]$$

- $\bar{V}_I$ estimable up to $\boldsymbol{\theta}$ given $\hat{\mathbb{P}}(\alpha^I(\boldsymbol{\xi},\rho) = 1 \mid \boldsymbol{\xi})$, $\hat{F}^\sigma(\xi'_j \mid \boldsymbol{\xi})$, and $\hat{\sigma}^x(\boldsymbol{\xi},\nu)$.
- $\sigma^x(\boldsymbol{\xi},\nu) = F_X^{-1}(1 - F_\nu(\nu) \mid \boldsymbol{\xi})$.
- Similar calculations in Jofre-Bonet and Pesendorfer (2003) and Pakes et al. (2007).

Back to integrated VF. Back to semi-parametric estimation.

Computational Considerations

- $[I - \beta \boldsymbol{M}(\boldsymbol{P})]\bar{\boldsymbol{V}}_I = \boldsymbol{\pi} - \boldsymbol{K}(\boldsymbol{\theta}_x) + \boldsymbol{\Sigma}(\boldsymbol{\theta}_\rho)$ must be solved for each $\boldsymbol{\theta}$.
- *However*, the matrix $I - \beta \boldsymbol{M}(\boldsymbol{P})$ is *banded*:

- The LU decomposition of a banded matrix has banded components.
  - Speeds up computation considerably.
  - Decomposition computed once and stored in memory.

Back to Monte Carlo Setup.

Bajari et al. (2007): Non-Linearity in $\boldsymbol{\theta}$

Note the model is not linear in parameters:

$$\bar{V}_l(\boldsymbol{\xi}) = \int_{\rho} \max\left\{\pi(\boldsymbol{\xi}) + \rho, \int V_l^A(\boldsymbol{\xi}, \nu)\mathrm{d}F_\nu\right\}\mathrm{d}F_\rho$$

Back to BBL.

Bajari et al. (2007): Non-Linearity in $\boldsymbol{\theta}$

Note the model is not linear in parameters:

$$\bar{V}_I(\boldsymbol{\xi}) = \pi(\boldsymbol{\xi}) + \int_{\rho} \max\left\{\rho, \int V_I^A(\boldsymbol{\xi}, \nu)\mathrm{d}F_\nu - \pi(\boldsymbol{\xi})\right\}\mathrm{d}F_\rho$$

Back to BBL.

Bajari et al. (2007): Non-Linearity in $\boldsymbol{\theta}$

Note the model is not linear in parameters:

$$\bar{V}_I(\boldsymbol{\xi}) = \pi(\boldsymbol{\xi}) + \int_{\int V_I^A(\boldsymbol{\xi},\nu)\mathrm{d}F_\nu - \pi(\boldsymbol{\xi})}^{\infty} \rho \mathrm{d}F_\rho + F_\rho\left(\int V_I^A(\boldsymbol{\xi},\nu)\mathrm{d}F_\nu - \pi(\boldsymbol{\xi})\right)\left[\int V_I^A(\boldsymbol{\xi},\nu)\mathrm{d}F_\nu - \pi(\boldsymbol{\xi})\right]$$

Back to BBL.

Bajari et al. (2007): Non-Linearity in $\boldsymbol{\theta}$

Note the model is not linear in parameters:

$$\bar{V}_I(\boldsymbol{\xi}) = \pi(\boldsymbol{\xi}) + \int_{F_\rho^{-1}[\mathbb{P}(\alpha^I(\boldsymbol{\xi},\rho)=1|\boldsymbol{\xi})]}^{\infty} \rho \mathrm{d}F_\rho + \mathbb{P}\big(\alpha^I(\boldsymbol{\xi},\rho)=1 \mid \boldsymbol{\xi}\big) F_\rho^{-1}\big[\mathbb{P}\big(\alpha^I(\boldsymbol{\xi},\rho)=1 \mid \boldsymbol{\xi}\big)\big]$$

$\int_a^\infty \rho \mathrm{d}F_\rho$ is not generally linear in parameters (e.g. exponential is, lognormal isn't).

Back to BBL.

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$\int_a^\infty \rho \mathrm{d}F_\rho$ is not generally linear in parameters (e.g. exponential is, lognormal isn't).

Nonetheless, forward simulation can still be performed once.

- Draw $u \sim U[0,1]$
- Exit: $\mathbb{1}\{u \leq \mathbb{P}(\alpha^I(\boldsymbol{\xi},\rho)=1 \mid \boldsymbol{\xi})\} \Leftrightarrow \mathbb{1}\{F_\rho^{-1}(u) \leq \int V_I^A(\boldsymbol{\xi},\nu)\mathrm{d}F_\nu - \pi(\boldsymbol{\xi})\}$

Back to BBL.

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Note the model is not linear in parameters:

$$\bar{V}_I(\boldsymbol{\xi}) = \pi(\boldsymbol{\xi}) + \int_{F_\rho^{-1}[\mathbb{P}(\alpha^I(\boldsymbol{\xi},\rho)=1|\boldsymbol{\xi})]}^{\infty} \rho \mathrm{d}F_\rho + \mathbb{P}(\alpha^I(\boldsymbol{\xi},\rho)=1 \mid \boldsymbol{\xi}) F_\rho^{-1}\left[\mathbb{P}(\alpha^I(\boldsymbol{\xi},\rho)=1 \mid \boldsymbol{\xi})\right]$$

$\int_a^\infty \rho \mathrm{d}F_\rho$ is not generally linear in parameters (e.g. exponential is, lognormal isn't).

Nonetheless, forward simulation can still be performed once.

- Draw $u \sim U[0,1]$
- Exit: $\mathbb{1}\{u \leq \mathbb{P}(\alpha^I(\boldsymbol{\xi},\rho)=1 \mid \boldsymbol{\xi})\} \Leftrightarrow \alpha^I(\boldsymbol{\xi}, F_\rho^{-1}(u))$
- Forward simulate, storing exit u_{τ_e}. As $\boldsymbol{\theta}_\rho$ changes, only recompute $F_\rho^{-1}(u_{\tau_e}; \boldsymbol{\theta}_\rho)$.
- Generalizes to investment, entry choices if wanted.

Back to BBL.

Flow Profit Specification: BBL Design

Demand

- Simple logit:

$$u_{ij} = \begin{cases} \varepsilon_{i0} & \text{if } j = 0 \\ \gamma_0 \xi_j + \alpha p_j + \varepsilon_{ij} & \text{if } j = 1, \ldots, N \end{cases},$$

- $\varepsilon_{ij} \sim T1EV$ iid across i and j.

Pricing

- Constant marginal cost: $\mu(\xi_j) = \exp(\theta_{c1} + \theta_{c2}\xi_j)$ with $\theta_{c2} = 0$.
- Firms compete à la Bertrand.
  - Caplin and Nalebuff (1991) $\Rightarrow \exists!$ Eqm $\Rightarrow \pi_j(\boldsymbol{\xi})$ well-defined.

Back to Monte Carlo setup.

Monte Carlo: First Stage and Deviation Details — BBL Design

- Empirical policy functions:

$$\hat{\sigma}^x(\boldsymbol{\xi}) = f^x(\boldsymbol{\xi})^\top \hat{\chi}^x \qquad \hat{\mathbb{P}}^E(\boldsymbol{\xi}_{-1}) = \Lambda\big(f^E(\boldsymbol{\xi})^\top \hat{\chi}^E\big) \qquad \hat{\mathbb{P}}^I(\boldsymbol{\xi}) = \Lambda\big(f^I(\boldsymbol{\xi})^\top \hat{\chi}^I\big).$$

- BBL deviations: asymptotic,

We draw $\tilde{\boldsymbol{\chi}}_i \sim \mathrm{N}(\hat{\boldsymbol{\chi}}, \hat{\boldsymbol{\Sigma}}_{\boldsymbol{\chi}})$ for each deviation i. We then form deviations as

$$\tilde{\sigma}_i^x(\boldsymbol{\xi}) = f^x(\boldsymbol{\xi})^\top \tilde{\chi}_i^x; \quad \tilde{\mathbb{P}}_i^E(\boldsymbol{\xi}_{-1}) = \Lambda\big(f^E(\boldsymbol{\xi})^\top \tilde{\chi}_i^E\big); \quad \tilde{\mathbb{P}}_i^I(\boldsymbol{\xi}) = \Lambda\big(f^I(\boldsymbol{\xi})^\top \tilde{\chi}_i^I\big).$$

Back to Monte Carlo Results.

Monte Carlo: First Stage and Deviation Details — BBL Design

- Empirical policy functions:

$$\hat{\sigma}^x(\boldsymbol{\xi}) = f^x(\boldsymbol{\xi})^\top \hat{\chi}^x \qquad \hat{\mathbb{P}}^E(\boldsymbol{\xi}_{-1}) = \Lambda\big(f^E(\boldsymbol{\xi})^\top \hat{\chi}^E\big) \qquad \hat{\mathbb{P}}^I(\boldsymbol{\xi}) = \Lambda\big(f^I(\boldsymbol{\xi})^\top \hat{\chi}^I\big).$$

- BBL deviations: multiplicative,

We draw $\iota_i \in \{.95, .975, 1.025, 1.05, 1.075\}$ for each deviation i. We then form deviations as

$$\tilde{\sigma}_i^x(\boldsymbol{\xi}) = \iota_i f^x(\boldsymbol{\xi})^\top \hat{\chi}^x; \quad \tilde{\mathbb{P}}_i^E(\boldsymbol{\xi}_{-1}) = \Lambda\big(\iota_i f^E(\boldsymbol{\xi})^\top \hat{\chi}^E\big); \quad \tilde{\mathbb{P}}_i^I(\boldsymbol{\xi}) = \Lambda\big(\iota_i f^I(\boldsymbol{\xi})^\top \hat{\chi}^I\big) \ ,$$

Back to Monte Carlo Results.

Monte Carlo: First Stage and Deviation Details — BBL Design

- Empirical policy functions:

$$\hat{\sigma}^x(\boldsymbol{\xi}) = f^x(\boldsymbol{\xi})^\top \hat{\chi}^x \qquad \hat{\mathbb{P}}^E(\boldsymbol{\xi}_{-1}) = \Lambda\big(f^E(\boldsymbol{\xi})^\top \hat{\chi}^E\big) \qquad \hat{\mathbb{P}}^I(\boldsymbol{\xi}) = \Lambda\big(f^I(\boldsymbol{\xi})^\top \hat{\chi}^I\big).$$

- BBL deviations: and additive.

We draw $o_{is} \sim N(0, 0.5)$ for each deviation i and simulated decision s.
We then form deviations as

$$\tilde{\sigma}^x_{is}(\boldsymbol{\xi}) = f^x(\boldsymbol{\xi})^\top \hat{\chi}^x + o^x_{is}; \quad \tilde{\mathbb{P}}^E_{is}(\boldsymbol{\xi}_{-1}) = \Lambda\big(f^E(\boldsymbol{\xi})^\top \hat{\chi}^E + o^E_{is}\big); \quad \tilde{\mathbb{P}}^I_{is}(\boldsymbol{\xi}) = \Lambda\big(f^I(\boldsymbol{\xi})^\top \hat{\chi}^I + o^I_{is}\big).$$

Back to Monte Carlo Results.

Flow Profit Specification

Demand

- Nested Logit:

$$u_{ij} = \begin{cases} \epsilon_i^{\text{out}} + (1-\varsigma)\varepsilon_{ij} & \text{if } j = 0 \\ \alpha p_j + \xi_j + \epsilon_i^{\text{in}} + (1-\varsigma)\varepsilon_{ij} & \text{if } j = 1, \ldots, N \end{cases},$$

- Inside and Outside good nests; $\epsilon_i^{\text{g}} + (1-\varsigma)\varepsilon_{ij} \sim T1EV, \text{g} = \{\text{in, out}\}$.

Pricing

- Marginal cost: $\mu(\xi_j) = \exp(\theta_{c1} + \theta_{c2}\xi_j)$.
- Firms compete à la Bertrand
  - Caplin and Nalebuff (1991) $\Rightarrow \exists!$ Eqm $\Rightarrow \pi_j(\boldsymbol{\xi})$ well-defined.

Back to Monte Carlo setup.

Monte Carlo: First Stage and Deviation Details

- Empirical policy functions:

$$\hat{\sigma}^x(\boldsymbol{\xi}, \nu_\tau) = f^x(\boldsymbol{\xi})^\top \hat{\chi}^x_\tau \qquad \hat{\mathbb{P}}^E(\boldsymbol{\xi}_{-1}) = \Lambda\big(f^E(\boldsymbol{\xi})^\top \hat{\chi}^E\big) \qquad \hat{\mathbb{P}}^I(\boldsymbol{\xi}) = \Lambda\big(f^I(\boldsymbol{\xi})^\top \hat{\chi}^I\big).$$

- BBL deviations: asymptotic,

We draw $\tilde{\boldsymbol{\chi}}_i \sim \mathrm{N}(\hat{\boldsymbol{\chi}}, \hat{\boldsymbol{\Sigma}}_{\boldsymbol{\chi}})$ for each deviation i. We then form deviations as

$$\tilde{\sigma}^x_i(\boldsymbol{\xi}, \nu_\tau) = f^x(\boldsymbol{\xi})^\top \tilde{\chi}^x_{i\tau}; \quad \tilde{\mathbb{P}}^E_i(\boldsymbol{\xi}_{-1}) = \Lambda\big(f^E(\boldsymbol{\xi})^\top \tilde{\chi}^E_i\big); \quad \tilde{\mathbb{P}}^I_i(\boldsymbol{\xi}) = \Lambda\big(f^I(\boldsymbol{\xi})^\top \tilde{\chi}^I_i\big).$$

Back to Monte Carlo Results.

Monte Carlo: First Stage and Deviation Details

- Empirical policy functions:

$$\hat{\sigma}^x(\boldsymbol{\xi}, \nu_\tau) = f^x(\boldsymbol{\xi})^\top \hat{\chi}^x_\tau \qquad \hat{\mathbb{P}}^E(\boldsymbol{\xi}_{-1}) = \Lambda\big(f^E(\boldsymbol{\xi})^\top \hat{\chi}^E\big) \qquad \hat{\mathbb{P}}^I(\boldsymbol{\xi}) = \Lambda\big(f^I(\boldsymbol{\xi})^\top \hat{\chi}^I\big).$$

- BBL deviations: multiplicative,

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Back to Monte Carlo Results.

Monte Carlo: First Stage and Deviation Details

- Empirical policy functions:

$$\hat{\sigma}^x(\boldsymbol{\xi}, \nu_\tau) = f^x(\boldsymbol{\xi})^\top \hat{\chi}^x_\tau \qquad \hat{\mathbb{P}}^E(\boldsymbol{\xi}_{-1}) = \Lambda\big(f^E(\boldsymbol{\xi})^\top \hat{\chi}^E\big) \qquad \hat{\mathbb{P}}^I(\boldsymbol{\xi}) = \Lambda\big(f^I(\boldsymbol{\xi})^\top \hat{\chi}^I\big).$$

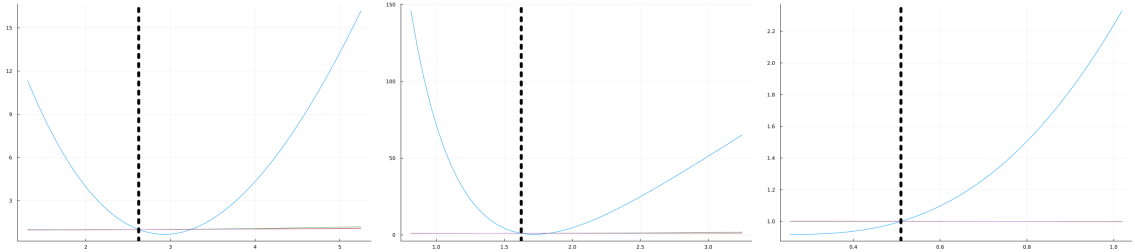
- BBL deviations: and additive.

We draw $o_{is} \sim N(0, 0.5)$ for each deviation i and simulated decision s.
We then form deviations as

$$\tilde{\sigma}^x_{is}(\boldsymbol{\xi}, \nu_\tau) = f^x(\boldsymbol{\xi})^\top \hat{\chi}^x_\tau + o^x_{is}; \quad \tilde{\mathbb{P}}^E_{is}(\boldsymbol{\xi}_{-1}) = \Lambda\big(f^E(\boldsymbol{\xi})^\top \hat{\chi}^E + o^E_{is}\big); \quad \tilde{\mathbb{P}}^I_{is}(\boldsymbol{\xi}) = \Lambda\big(f^I(\boldsymbol{\xi})^\top \hat{\chi}^I + o^I_{is}\big).$$

Back to Monte Carlo Results.

Objective Function: Investment Cost Parameters



θ_{x1} θ_{x2} θ_{x3}

$$c(x, \nu) = \theta_{x1} x + \theta_{x2} x^2 + \theta_{x3} x \nu$$

Indirect Inference | BBL - Asymptotic | BBL - Multiplicative | BBL - Additive

- Note: legacy results based on Indirect Inference.
  - Focus: shape of BBL objectives vs. the recursive estimator's.

Back to Monte Carlo Results.

Semi-parametric Estimation

☠ This is very preliminary.

Model + data + $\boldsymbol{\theta}$ induce an operator $T_{\boldsymbol{\theta}} : \mathcal{F}_\nu \to \mathcal{F}_\nu$ as follows:

$$F_\nu \mapsto \sigma^x(\boldsymbol{\xi}, \nu) \mapsto \bar{V}_I \mapsto W(\xi' \mid \boldsymbol{\xi}, F^\sigma) \mapsto F_\nu$$

▸ Integrated Value Function Details

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Then, we can estimate $\boldsymbol{\theta}$ by solving

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- In the example below, $\sum_\xi d\left(\hat{F}_X(\cdot \mid \xi), F_X(\cdot \mid \xi, \boldsymbol{\theta})\right)$, where $d(F, G) = \|F^{-1} - G^{-1}\|_\infty$

Semi-parametric Estimation: Example

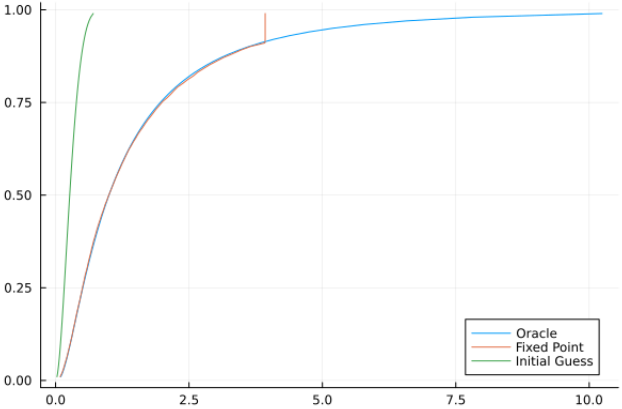
We test the feasibility of the idea above in a very simple setting.

- A monopolist solving a dynamic decision problem.
- $D(p;\xi) = \xi - p \Rightarrow \pi(\xi)$.
- $\xi \in \{\xi_L, \xi_H\}$.
- Monopolist controls dynamics of ξ via $P(\xi_L \mid \xi, x)$.

This is a <u>rather difficult</u> DGP:

- Only two states.
- Stationary distribution has $P(\xi_H) = 0.9$.
- And still ...

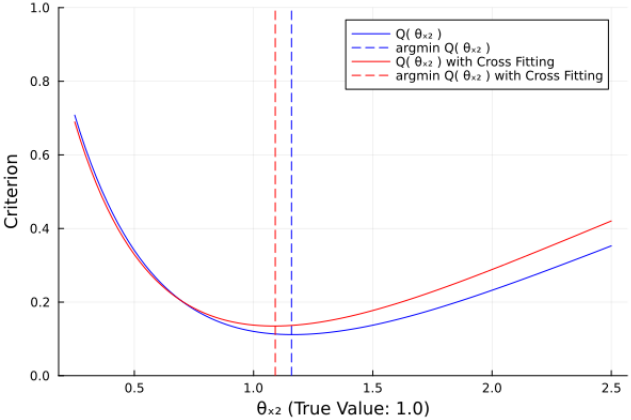
Semi-parametric Estimation: T_θ



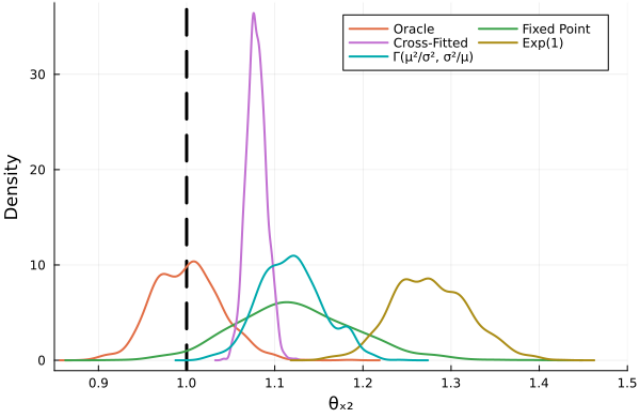
T_θ seems rather stable in this example.

- Looks like a contraction with a rather low contraction modulus.
- We have found convergence for all $\boldsymbol{\theta}$ and F_ν^0.

Semi-parametric Estimation: Example



Semi-parametric Estimation: Monte Carlo with Misspecification



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